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First-Line Sugemalimab Plus Chemotherapy for Advanced Gastric Cancer
The GEMSTONE-303 Randomized Clinical Trial

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IMPORTANCE Gastric cancer, including gastroesophageal junction cancer, is one of the most commonly diagnosed cancers worldwide, with high mortality. Sugemalimab is a fully human anti–programmed death-ligand 1 (PD-L1) antibody. The combination of sugemalimab and chemotherapy showed promising antitumor activity and safety in a phase 1b study among patients with treatment-naive, unresectable, locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma. This combination was further evaluated in the GEMSTONE-303 phase 3 trial.

OBJECTIVE To evaluate the efficacy of sugemalimab in combination with capecitabine and oxaliplatin (CAPOX) compared with placebo plus CAPOX as first-line treatment for patients with unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma with PD-L1 combined positive score (CPS) of 5 or greater.

DESIGN, SETTING, AND PARTICIPANTS GEMSTONE 303 is a phase 3, randomized, double-blind, placebo-controlled study conducted at 54 sites in China that enrolled patients from April 9, 2019, through December 29, 2021, with follow-up to July 9, 2023. A total of 479 eligible patients with unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma with PD-L1 CPS of 5 or greater who did not receive any prior systemic therapy were randomized.

INTERVENTION Patients received sugemalimab (1200 mg intravenously) (n = 241) or placebo (n = 238) every 3 weeks for up to 24 months, plus CAPOX every 3 weeks for up to 6 cycles.

MAIN OUTCOMES AND MEASURES Primary outcomes were overall survival and investigator-assessed progression-free survival.

RESULTS Baseline characteristics were well balanced between the 2 groups. Most patients were male (71.4% in sugemalimab group, 74.8% in placebo group). Median follow-up was 25.1 months in the sugemalimab group and 26.3 months in the placebo group. The sugemalimab group demonstrated significant improvements in overall survival (median, 15.6 months [95% CI, 13.3-17.8] vs 12.6 months [95% CI, 10.6-14.1]; hazard ratio, 0.75 [95% CI, 0.61-0.92]; $P = .006$) and progression-free survival (median, 7.6 months [95% CI, 6.4-7.9] vs 6.1 months [95% CI, 5.1-6.4]; hazard ratio, 0.66 [95% CI, 0.54-0.81]; $P < .001$). Grade 3 or higher treatment-related adverse events occurred in 53.9% of patients in the sugemalimab group and 50.6% in the placebo group.

CONCLUSIONS AND RELEVANCE Sugemalimab plus chemotherapy significantly prolonged overall survival and progression-free survival with a manageable safety profile in previously untreated patients with unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma.

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Gastric cancer, including gastroesophageal junction cancer, is the fifth most commonly diagnosed cancer and the fifth leading cause of cancer-related deaths worldwide.[1] The incidence is higher in East Asia compared with other regions,[2] particularly in China.[3] Adenocarcinoma accounts for more than 90% of all cases. Before availability of programmed death 1 (PD-1) inhibitors, the standard first-line therapy for unresectable locally advanced or metastatic gastric or gastroesophageal junction cancer was fluoropyrimidine plus platinum-based chemotherapy, with a poor median overall survival of 1 year or less.[4-8]

Adding a PD-1 inhibitor to doublet chemotherapy extended overall survival by 2 to 4 months, establishing it as the new standard first-line therapy.[6,9-13] This benefit was demonstrated in several phase 3 clinical trials of anti-PD-1 antibodies, namely nivolumab (CheckMate-649), pembrolizumab (KEYNOTE-859), sintilimab (ORIENT-16), and tislelizumab (RATIONALE-305).[6,10,11,13] However, 16% to 36% of patients discontinued due to adverse events, with 27% to 31% experiencing immune-related events, of which 5% to 8% were grade 3 and above.[6,10,11,13]

The programmed death-ligand 1 (PD-L1) is a co-inhibitory factor of the immune response.[14] Blocking PD-L1 has a similar effect on the PD-1/PD-L1 axis as anti-PD-1 agents but may result in less toxicity by sparing programmed death-ligand 2.[15] However, evidence supporting PD-L1 inhibitors in advanced gastric cancer remains limited. Baseline PD-L1 expression, irrespective of cutoff values (eg, combined positive score [CPS] ≥1, ≥5, or ≥10), has been associated with improved survival outcomes,[6,10,11,13] consistent with findings in other tumor types.[16,17] However, prior trials of anti-PD-1 antibodies enrolled patients regardless of PD-L1 expression levels.

Sugemalimab is a full-length, fully human immunoglobulin G4 (s228p) monoclonal antibody targeting PD-L1 that retains FcγRI (a receptor for the Fc region of IgG) binding, enabling antibody-dependent cellular phagocytosis via cross-linking of PD-L1–expressing tumor cells with macrophages.[18] In the GEMSTONE-101 phase 1b study, sugemalimab with capecitabine plus oxaliplatin (CAPOX) demonstrated good tolerability and encouraging efficacy in 29 patients with gastric or gastroesophageal junction adenocarcinoma, achieving an objective response rate (ORR) of 62.1%, median progression-free survival of 8.3 months, and median overall survival of 17.0 months. Subgroup analyses suggested better outcomes in patients with higher CPS (≥5 vs <5).[19]

Nivolumab, the first anti-PD-1 agent investigated in first-line gastric cancer, was not approved until 2021. To build on previous findings and differentiate from other phase 3 trials of checkpoint inhibitors, GEMSTONE-303 was initiated in 2019, assessing sugemalimab as the first anti-PD-L1 agent in this setting. GEMSTONE-303 is also the first trial to selectively enroll patients with CPS 5 or greater.

Key Points

Question Does adding sugemalimab to chemotherapy improve survival for untreated advanced gastric and gastroesophageal junction adenocarcinoma with programmed death-ligand 1 (PD-L1) combined positive score (CPS) of 5 or more?

Findings In this randomized clinical trial of 479 patients with PD-L1 CPS of 5 or more, sugemalimab plus chemotherapy significantly prolonged overall survival (median, 15.6 vs 12.6 months) and progression-free survival (median, 7.6 vs 6.1 months) compared with chemotherapy alone, with a manageable safety profile.

Meaning GEMSTONE-303 is the first trial demonstrating a statistically significant survival benefit for addition of an anti–PD-L1 antibody to standard chemotherapy for advanced gastric and gastroesophageal junction adenocarcinoma.

Methods

Study Design and Participants

GEMSTONE-303 was a randomized, double-blind, placebo-controlled, phase 3 study conducted at 54 sites in China. Eligible patients were aged 18 to 75 years; with unresectable, locally advanced or metastatic gastric or gastroesophageal junction cancer; with histologically confirmed adenocarcinoma; with adequate tumor tissue for PD-L1 assessment; PD-L1 CPS 5 or greater at baseline; and with at least 1 measurable or evaluable lesion according to Response Evaluation Criteria in Solid Tumours (RECIST), version 1.1. Key exclusion criteria included known positive human epidermal growth factor receptor 2 (*HER2*; also known as *ERBB2*) status, prior systemic therapy, and disease progression within 6 months of adjuvant or neoadjuvant chemotherapy. Full eligibility criteria are available in the protocol (Supplement 1).

The trial adhered to the Declaration of Helsinki, Good Clinical Practice Guideline, and applicable local laws and regulations. The protocol was approved by the independent ethics committee at each site. All patients provided written informed consent. An independent data monitoring committee oversaw safety and efficacy. This study followed the Consolidated Standards of Reporting Trials reporting guideline.

Randomization and Masking

Eligible patients were randomized 1:1 to receive either sugemalimab or placebo plus CAPOX chemotherapy. Randomization was performed using an interactive web response system, stratified by PD-L1 CPS (5-9 or ≥10) and Eastern Cooperative Oncology Group performance status score (0 or 1). Patients, investigators, and the sponsor were masked to treatment allocation and PD-L1 expression levels.

Treatment and Assessment

Patients received sugemalimab (1200 mg intravenously) or placebo every 21 days for up to 24 months, plus CAPOX (oral capecitabine 1000 mg/m^2/d twice daily on days 1-14 and intravenous oxaliplatin 130 mg/m^2 on day 1) every 21 days for up to 6 cycles. Treatment continued until disease progression, unacceptable toxicity, consent withdrawal, or other protocol-specified reasons. Postprogression treatment was allowed at the investigator's discretion.

Tumor response was assessed per RECIST version 1.1 by the investigator at baseline, week 9, then every 6 weeks for the first

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year, and every 12 weeks thereafter. Following treatment discontinuation, survival was assessed every 12 weeks. Adverse events were evaluated throughout the study and for up to 90 days from the last dose of study treatment, graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.03, and summarized based on the Medical Dictionary for Regulatory Activities, version 24.1. Immune-related adverse events (irAEs) were assessed by the sponsor. PD-L1 expression was evaluated centrally by immunohistochemistry (Ventana SP263 antibody) on the Ventana BenchMark ULTRA automated platform (Roche Tissue Diagnostics), with CPS determined by the percentage of positive tumor cells and mononuclear inflammatory cells among the total viable tumor cells.

Outcomes

The dual primary end points were overall survival and investigator-assessed progression-free survival in all randomized patients. Key secondary end points included progression-free survival assessed by blinded independent central review committee, ORR, and duration of response. Exploratory analyses examined the association between PD-L1 expression and efficacy. Further outcome details are available in the protocol (Supplement 1).

Statistical Analysis

All analyses followed the statistical analysis plan. The overall type I error (2-sided α = .05) was controlled using weighted Bonferroni and Fallback method. When the progression-free survival analysis was statistically significant, the significance level of α = .01 would be recycled to overall survival analysis. With 480 patients, 360 progression-free survival events among all randomized patients were estimated to provide 83% power to detect a hazard ratio (HR) of 0.69 at a 2-sided α level of .01, and 360 overall survival events among all randomized patients would provide 85% power to detect an HR of 0.72 at a 2-sided α level of .04. An interim analysis and final analysis were planned for overall survival. Type I errors were controlled using a sequential testing approach and the O'Brien-Fleming boundary based on the Lan-DeMets method. The interim analysis (final progression-free survival and interim overall survival) occurred at 270 deaths.

Overall survival and progression-free survival were compared using a stratified log-rank test. Hazard ratios with 95% CIs were estimated using the stratified Cox proportional-hazards model, based on randomization stratification factors. Kaplan-Meier curves and median survival were estimated for each treatment group. The stratified Mantel-Haenszel test was used for comparisons of ORR. Subgroup analyses of progression-free survival and overall survival were assessed descriptively by HR (95% CIs) based on unstratified Cox model. More details are reported in Supplement 2.

Results

Patients

Between April 2019 and December 2021, a total of 1227 patients were screened for eligibility at 54 sites in China, with 479 randomized to sugemalimab (241 patients) or placebo (238 patients) (intent-to-treat [ITT] population). Safety population included 241 and 237 patients who received at least 1 dose of sugemalimab or placebo plus CAPOX, respectively (**Figure 1**).

Baseline characteristics were well balanced (**Table 1**). All patients were Asian. Most patients were male (172 [71.4%] patients in the sugemalimab group and 178 [74.8%] in the placebo group) and had an Eastern Cooperative Oncology Group performance status of 1 (179 [74.3%] vs 182 [76.5%]). There were 130 patients (53.9%) in the sugemalimab group and 128 (53.8%) in the placebo group who had PD-L1 CPS 10 or greater.

In the ITT population, 129 patients (53.5%) in the sugemalimab group and 145 (60.9%) in the placebo group received subsequent anticancer therapy (eTable 1 in Supplement 2). Forty-three (17.8%) patients and 56 (23.5%) patients, respectively, received subsequent PD-1 or PD-L1 inhibitor.

Primary Outcomes

The final overall survival analysis was performed as of July 9, 2023, with a median follow-up of 25.1 months (IQR, 21.22-34.14) in the sugemalimab group and 26.3 months (IQR, 21.19-44.06) in the placebo group. A total of 212 patients (88.0%) in the sugemalimab group and 227 (95.4%) in the placebo group discontinued treatment, mostly due to radiographic disease progression (140 [58.1%] vs 170 [71.4%]).

At the final analysis for overall survival, 170 deaths (70.5%) were observed in the sugemalimab group vs 190 (79.8%) in the placebo group. Median overall survival was 15.6 months (95% CI, 13.3-17.8) in the sugemalimab group compared with 12.6 months (95% CI, 10.6-14.1) in the placebo group (stratified HR, 0.75 [95% CI, 0.61-0.92]; P = .006) (**Figure 2**A; eFigure 1A and eTable 2 in Supplement 2). Sugemalimab plus CAPOX showed superior overall survival vs CAPOX alone, with a 25% reduction in the risk of death.

At the final analysis for progression-free survival (data cutoff date: August 6, 2022), 177 (73.4%) patients and 201 (84.5%) patients experienced disease progression or death as assessed by the investigator in the 2 groups. Median progression-free survival was 7.6 months (95% CI, 6.4-7.9) vs 6.1 months (95% CI, 5.1-6.4) (stratified HR, 0.66 [95% CI, 0.54-0.81]; P < .001) (Figure 2B; eFigure 1B and eTable 2 in Supplement 2).

Secondary Outcomes

Sugemalimab plus CAPOX significantly prolonged overall survival in patients with PD-L1 CPS 10 or greater; median overall survival was 17.8 months (95% CI, 14.4-21.2) in the sugemalimab group, compared with 12.5 months (95% CI, 9.9-13.9) in the placebo group (stratified HR, 0.64 [95% CI, 0.48-0.85]; P = .002) (Figure 2C; eFigure 1C and eTable 2 in Supplement 2). In patients with PD-L1 CPS 10 or greater, the sugemalimab group showed a median progression-free survival of 7.8 months (95% CI, 6.8-10.0) compared with 5.5 months (95% CI, 4.8-6.7) (stratified HR, 0.58 [95% CI, 0.43-0.77]; P < .001) (Figure 2D; eFigure 1D and eTable 2 in Supplement 2).

Subgroup analyses of investigator-assessed overall survival and progression-free survival in the ITT population

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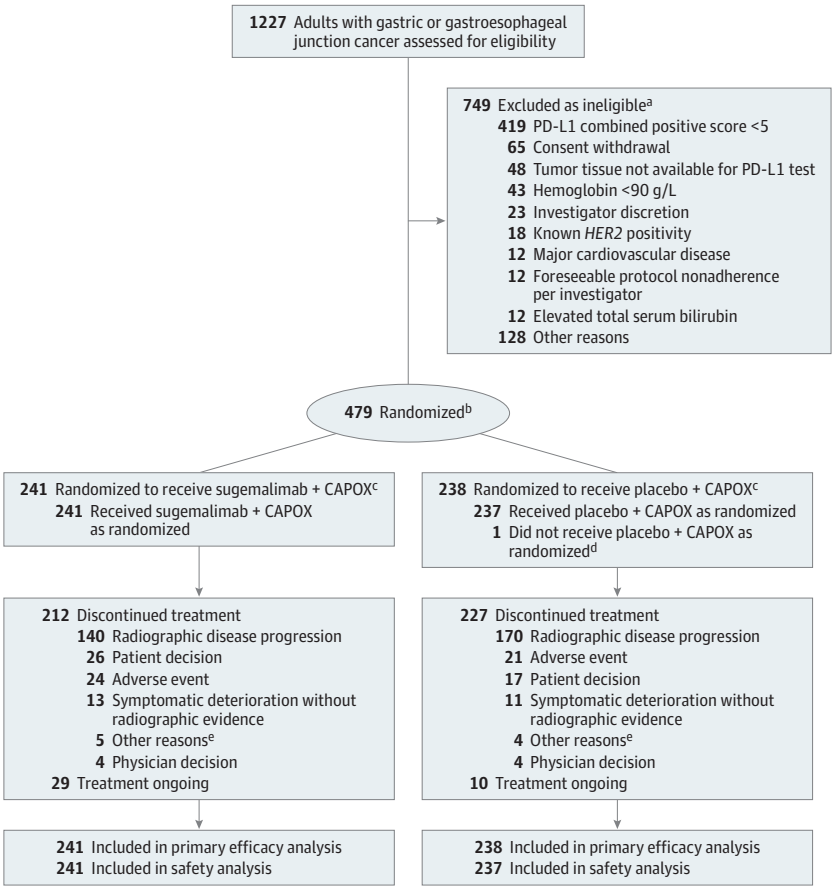
Figure 1. Patient Flow in the GEMSTONE-303 Trial



CAPOX indicates capecitabine and oxaliplatin.

[a]Some patients met more than 1 ineligibility criterion during screening.

[b]Eligible patients were randomized in a 1:1 ratio and stratified by programmed death-ligand 1 (PD-L1) expression (combined positive score [CPS], 5-9 or ≥10) and Eastern Cooperative Oncology Group (ECOG) performances status score (0 or 1). PD-L1 expression was measured by CPS, which was defined as the number of PD-L1–staining cells (tumor cells, lymphocytes, and macrophages) divided by the total number of viable tumor cells present in the sample multiplied by 100. The maximum score is defined as 100 when the calculation exceeds 100. PD-L1 expression was assessed at a central laboratory by immunohistochemistry using the Ventana PD-L1 (SP263) assay on a BenchMark autostainer assay (Roche Tissue Diagnostics) according to the manufacturer's instructions. The ECOG performance status score 5-point scale is commonly used for selection of appropriate patients in clinical trials. A score of 0 indicates fully active, able to carry on all predisease performance without restriction; 1, restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature.

[c]CAPOX regimen was intravenous oxaliplatin (130 mg/m^2) on day 1 plus oral capecitabine (1000 mg/m^2/d twice daily) on days 1-14, administered every 21 days.

[d]One patient in the placebo group did not receive any study treatment because the patient was found to have symptoms related to brain metastasis after randomization; therefore, this patient was included in the placebo group for the intent-to-treat population and was excluded from the safety analysis set.

[e]Other reasons for discontinuation: 5 patients in the sugemalimab group discontinued treatment for subsequent surgery (n = 3), subsequent radical therapy (n = 1), and drug interruption for more than 12 weeks due to adverse event (n = 1); 4 patients in the placebo group discontinued treatment for subsequent surgery (n = 3) and legal reasons (n = 1).

indicated that HRs favored sugemalimab in all prespecified subgroups (**Figure 3**; eFigure 2 in Supplement 2).

Progression-free survival results assessed by blinded independent central review committee were consistent with investigator assessments (PD-L1 CPS ≥5: stratified HR, 0.72; P = .002; PD-L1 CPS ≥10: stratified HR, 0.70; P = .02) (eFigure 3 in Supplement 2).

Best overall response is summarized in eTable 3 in Supplement 2. In patients with PD-L1 CPS 5 or greater, the sugemalimab group ORR was 68.6% compared with 52.7% in the placebo group, leading to a between-group difference of 15.9% (95% CI, 6.6%-25.2%; P = .001). Median duration of response was 6.9 months (95% CI, 5.6-8.4) vs 4.6 months (95% CI, 4.1-5.8) in patients with PD-L1 CPS 5 or greater and was 7.2 months (95% CI, 5.7-11.6) and 5.6 months (95% CI, 4.1-7.0) in patients with PD-L1 CPS 10 or greater (eTable 4 and eFigure 4 in Supplement 2).

Adverse Events

As of July 9, 2023, the median treatment duration was 6.3 months (range, 0.3-46.8) for sugemalimab and 5.2 months (range, 0.7-44.5) for placebo. Adverse event of any cause occurred in 240 patients (99.6%) in the sugemalimab group and 234 (98.7%) in the placebo group (eTable 5 in Supplement 2).

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The incidence of treatment-related adverse events of any grade were comparable between the 2 groups (98.8% vs 97.9%), with 130 patients (53.9%) and 120 (50.6%), respectively, reporting grade 3-5 events. The most common grade 3-5 treatment-related adverse event was decreased platelet count (44 [18.3%] vs 38 [16.0%]) (**Table 2**). Adverse events leading to discontinuation of any study treatment were observed in 43 (17.8%) and 29 (12.2%) patients, respectively. Treatment-related serious adverse events occurred in 80 patients (33.2%) in the sugemalimab group and 60 (25.3%) in the placebo group. The most common treatment-related serious adverse events were decreased platelet count (15 [6.2%] vs 16 [6.8%]), anemia (8 [3.3%] vs 5 [2.1%]), and decreased neutrophil count (6 [2.5%] vs 4 [1.7%]). Adverse events led to death of 7 (2.9%) patients in the sugemalimab group and 9 (3.8%) patients in the placebo group; 2 (0.8%) and 3 (1.3%) of the deaths, respectively, were caused by treatment-related adverse events.

Immune-related adverse events assessed by sponsor occurred in 54 patients (22.4%) and 12 (5.1%) in the 2 groups, in which 10 (4.1%) and 3 (1.3%) were grade 3 or greater, respectively (eTable 5 in Supplement 2). Most irAEs identified were grade 1-2, no fatal irAE was reported in either group, and no grade 4-5 irAEs such as pneumonitis or colitis occurred in the sugemalimab group (eTable 6 in Supplement 2). The most frequently reported irAEs included hypothyroidism (27 [11.2%] vs 3 [1.3%]), hyperthyroidism (15 [6.2%] vs 3 [1.3%]), and nonsevere skin adverse reactions (9 [3.7%] vs 1 [0.4%]).

Table 1. Baseline Characteristics of the Intent-to-Treat Population

Characteristic	No. (%)	
	Sugemalimab + CAPOX (n = 241)	Placebo + CAPOX (n = 238)
Age, median (range), y	63.0 (25-75)	63.0 (26-75)
Age group, y		
<65	137 (56.8)	135 (56.7)
≥65	104 (43.2)	103 (43.3)
Sex		
Male	172 (71.4)	178 (74.8)
Female	69 (28.6)	60 (25.2)
Baseline ECOG performance status[a]		
0	62 (25.7)	56 (23.5)
1	179 (74.3)	182 (76.5)
Primary tumor location		
Stomach	221 (91.7)	208 (87.4)
Gastroesophageal junction	20 (8.3)	30 (12.6)
No. of organs with metastasis		
1-2	132 (54.6)	132 (55.5)
≥3	90 (37.3)	91 (38.2)
Location of metastasis		
Liver	100 (41.5)	86 (36.1)
Retroperitoneal lymph node	93 (38.6)	89 (37.4)
Lung	41 (17.0)	53 (22.3)
Peritoneum	35 (14.5)	29 (12.2)
Tumor stage at screening		
III	9 (3.7)	10 (4.2)
IV	232 (96.3)	228 (95.8)
Prior cancer-related surgery		
Yes	60 (24.9)	55 (23.1)
No	181 (75.1)	183 (76.9)
PD-L1 expression (CPS)[b]		
5-9	111 (46.1)	110 (46.2)
≥10	130 (53.9)	128 (53.8)

Abbreviations: CAPOX, capecitabine plus oxaliplatin; CPS, combined positive score; ECOG, Eastern Cooperative Oncology Group; PD-L1, programmed death-ligand 1.

[a] The ECOG performance status score 5-point scale is commonly used for selection of appropriate patients in clinical trials. A score of 0 indicates fully active, able to carry on all predisease performance without restriction; 1, restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature.

[b] PD-L1 expression was measured by CPS, which was defined as the number of PD-L1–staining cells (tumor cells, lymphocytes, and macrophages) divided by the total number of viable tumor cells present in the sample multiplied by 100. The maximum score is defined as 100 when the calculation exceeds 100. PD-L1 expression was assessed at a central laboratory by immunohistochemistry using the Ventana PD-L1 (SP263) assay on a BenchMark autostainer assay (Roche Tissue Diagnostics) according to the manufacturer's instructions.

Discussion

GEMSTONE-303 is the first randomized, phase 3 study of a PD-L1 inhibitor in combination with chemotherapy showing significantly prolonged survival benefit in the first-line setting compared with chemotherapy alone in patients with unresectable locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma, for which PD-1 inhibitors have demonstrated efficacy. This is also the first to restrict enrollment to patients with CPS 5 or greater. The Kaplan-Meier curves of overall survival and progression-free survival illustrated an early separation that persisted throughout the evaluation period, suggesting sustained benefits. Subgroup analyses also indicated that HRs favored the sugemalimab group in all prespecified subgroups.

The predictive value of baseline PD-L1 expression to treatment effect of anti-PD-1- and anti-PD-L1-based therapies is widely recognized.[13,16,17,20] Different types of PD-L1 immunohistochemical assays and scoring algorithms are currently available, such as CPS and tumor proportion score.[16,17,21] GEMSTONE-303 also explored survival outcomes in this context. The HR observed for progression-free survival was 0.78 in patients with CPS 5 through 9 and 0.58 in those with CPS 10 or greater, and for overall survival, the HRs were 0.88 and 0.65, respectively (Figure 3; eFigure 2 in Supplement 2). Our data further confirm that higher CPS might be predictive of greater benefit of sugemalimab. Previous studies have shown that gastric or gastroesophageal junction cancer patients with low or negative PD-L1 expression gained little benefit from immunotherapy,[20,22] including patients with CPS less than 5 in ORIENT-16 and CheckMate-649.[6,11]

It is worth noting that patients with PD-L1 CPS 5 or greater were prospectively screened in GEMSTONE-303, while patients were enrolled regardless of baseline PD-L1 expression

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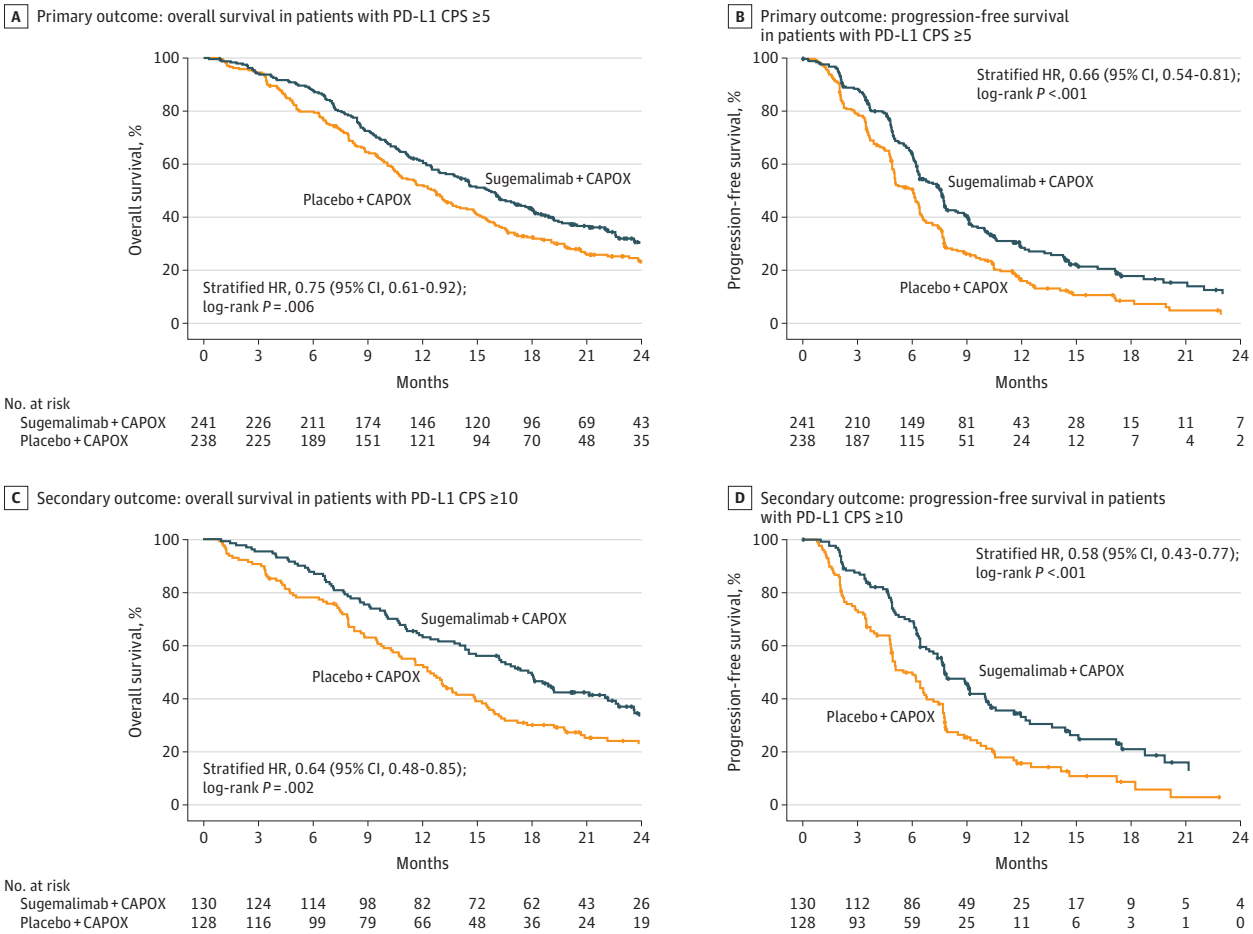
Figure 2. Estimates for Overall Survival and Progression-Free Survival



For all randomized patients, median follow-up time was 25.1 months (IQR, 21.22-34.14) for sugemalimab plus chemotherapy and 26.3 months (IQR, 21.19-44.06) for placebo plus chemotherapy. Programmed death-ligand 1 (PD-L1) expression was measured by combined positive score (CPS), which was defined as the number of PD-L1–staining cells (tumor cells, lymphocytes, and macrophages) divided by the total number of viable tumor cells present in the sample multiplied by 100. The maximum score is defined as 100 when the calculation exceeds 100. Data cutoff date for overall survival analysis was July 9, 2023. Data cutoff date for progression-free survival analysis was August 6, 2022. See eFigure 1 in Supplement 2 for the full survival curves. CAPOX indicates capecitabine and oxaliplatin; HR, hazard ratio.

in previous trials in the same setting. In GEMSTONE-303, among 1099 patients whose PD-L1 expression was evaluable at screening, the prevalence of PD-L1 CPS 5 or greater and CPS 10 or greater was 60% and 32%, respectively, which is in line with the percentages found in CheckMate-649 (60% for PD-L1 CPS ≥5), ORIENT-16 (61% for PD-L1 CPS ≥5), and KEYNOTE-859 (35% for PD-L1 CPS ≥10).[6,10,11]

Although ORIENT-16 included a similar percentage of patients with CPS 5 or greater as GEMSTONE-303 and CheckMate-649, patients enrolled in ORIENT-16 had slightly longer overall survival.[6,11] This difference is potentially due to the higher percentage of patients with CPS 10 or greater enrolled in ORIENT-16 than in GEMSTONE-303 (74.1% vs 53.9% among patients with CPS ≥5; calculated based on reported data).[11]

The design of the GEMSTONE-303 study carefully balanced the potential benefits of treatment with the risks of ongoing toxicity, setting a maximal tolerable duration of 6 cycles for CAPOX. This approach aligns with previous first-line chemotherapy studies for gastric cancer, in which the median number of capecitabine cycles ranged from 4 to 6[23-25]; also, 6 cycles was equivalent to the median durations of chemotherapy in CheckMate-649 and KEYNOTE-062 (4.9 and 4.7 months, respectively).[6,12] There is also no conclusive evidence from large randomized trials to suggest that maintenance therapy with capecitabine provides further overall survival benefit; therefore, GEMSTONE-303 was designed to stop both capecitabine and oxaliplatin after 6 cycles.

The treatment landscape is evolving with the increased utilization of anti-PD-1 and anti-PD-L1 antibodies. Earlier investigations have demonstrated overall survival benefits associated with later-line anti-PD-1 and anti-PD-L1 treatments for advanced gastric cancer.[20,22,26,27] In ATTRACTION-4, positive progression-free survival results failed to translate into overall survival benefits, perhaps due to confounding effects

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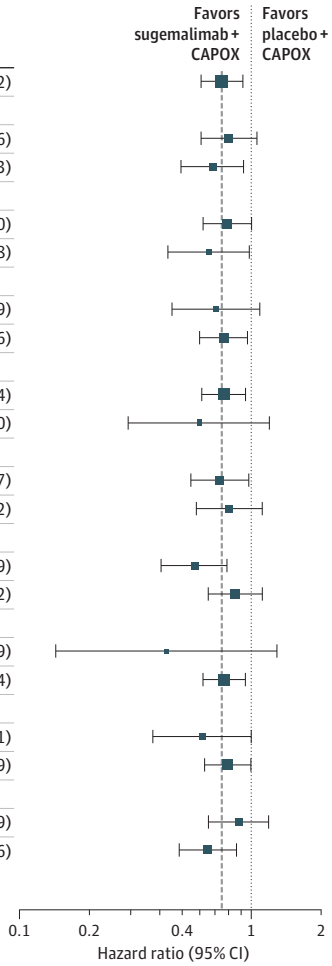
Figure 3. Analysis of Overall Survival in Prespecified Subgroups for Sugemalimab Plus CAPOX vs Placebo Plus CAPOX in the Intent-to-Treat Population

Baseline risk factor	Sugemalimab + CAPOX (n = 241)		Placebo + CAPOX (n = 238)		Hazard ratio (95% CI)
	No. of events/No. of participants	Median overall survival, mo	No. of events/No. of participants	Median overall survival, mo	
All patients	170/241	15.6	190/238	12.6	0.75 (0.61-0.92)
Age, y					
<65	97/137	14.9	102/135	12.6	0.80 (0.61-1.06)
≥65	73/104	16.0	88/103	12.8	0.68 (0.50-0.93)
Sex					
Male	125/172	16.0	140/178	12.7	0.79 (0.62-1.00)
Female	45/69	15.4	50/60	12.0	0.65 (0.44-0.98)
Baseline ECOG performance status					
0	42/62	16.3	39/56	13.1	0.70 (0.45-1.09)
1	128/179	14.6	151/182	12.2	0.76 (0.60-0.96)
Primary tumor location					
Stomach	157/221	14.6	167/208	12.5	0.76 (0.61-0.94)
Gastroesophageal junction	13/20	19.6	23/30	14.9	0.59 (0.29-1.20)
No. of organs with metastasis					
1-2	85/132	18.0	104/132	14.8	0.73 (0.55-0.97)
≥3	71/90	12.2	73/91	10.1	0.80 (0.58-1.12)
Liver metastases					
Yes	71/100	16.2	75/86	10.7	0.57 (0.41-0.79)
No	99/141	15.5	115/152	13.7	0.85 (0.65-1.12)
Tumor stage at screening					
III	5/9	26.8	9/10	12.5	0.43 (0.14-1.29)
IV	165/232	15.5	181/228	12.6	0.76 (0.62-0.94)
Prior treatment					
Yes	29/51	19.2	36/47	13.7	0.62 (0.38-1.01)
No	141/190	14.5	154/191	12.5	0.79 (0.63-0.99)
PD-L1 expression (CPS), %					
5-9	82/111	13.7	87/110	12.8	0.88 (0.65-1.19)
≥10	88/130	17.8	103/128	12.5	0.65 (0.49-0.86)



For all patients presented on the first row, hazard ratio and the associated confidence interval are calculated using stratified Cox regression model. For subgroup analyses, hazard ratios and the associated confidence intervals are calculated using unstratified Cox regression model. The vertical dashed line indicates the hazard ratio for all patients. The size of the symbol is proportional to the size of the population in the subgroup. For all randomized patients, median follow-up time was 25.1 months (IQR, 21.22-34.14) for sugemalimab plus chemotherapy and 26.3 months (IQR, 21.19-44.06) for placebo plus chemotherapy. CAPOX indicates capecitabine and oxaliplatin; CPS, combined positive score; ECOG, Eastern Cooperative Oncology Group; PD-L1, programmed death-ligand 1.

from subsequent immunotherapy after disease progression.[9] In this study, a higher proportion of patients received subsequent immunotherapy in the placebo group (23.5%) than in the sugemalimab group (17.8%) (eTable 1 in Supplement 2), which is notably higher than the percentages of patients receiving subsequent immunotherapy in CheckMate-649, KEYNOTE-859, and ORIENT-16 (8%-10.5%).[6,10,11] Despite this substantial and unbalanced receipt of subsequent immunotherapy, a statistically significant increase in overall survival was achieved, suggesting that the benefit was primarily derived from the first-line sugemalimab plus chemotherapy treatment.

The safety profile of sugemalimab plus chemotherapy was consistent with the known safety results of the individual treatments, and no new safety signals were identified.[23,28-31] In GEMSTONE-303, the most common adverse events reported in the sugemalimab and placebo groups were consistent with that of each agent and with the underlying disease. Adverse events leading to death occurred in less than 4% patients; specifically, only 1 patient's death was considered related to sugemalimab, and 2 deaths were related to placebo. It was also worth noting that no grade 4 or 5 irAEs, and only 4.1% grade 3 irAEs, were reported in the sugemalimab group. Overall, the data did not show additive toxicity when combining sugemalimab with chemotherapy, and the safety profile of this combination regimen was manageable.

Limitations

Several limitations of this study should be acknowledged. First, enrollment was limited to patients in China. Nonetheless, it is worth noting that CheckMate-649 demonstrated consistent efficacy between Chinese and global populations,[32]

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Table 2. Treatment-Related Adverse Events That Occurred in at Least 10% of Patients in All Patients Who Received Study Treatment

	No. (%)			
	Sugemalimab + CAPOX (n = 241)		Placebo + CAPOX (n = 237)	
Event	Grade 1-2[a]	Grade 3-4	Grade 1-2	Grade 3-4
Treatment-related adverse events[b]	108 (44.8)	128 (53.1)	112 (47.3)	117 (49.4)
Nausea	106 (44.0)	1 (0.4)	102 (43.0)	3 (1.3)
White blood cell count decreased	98 (40.7)	16 (6.6)	90 (38.0)	7 (3.0)
Anemia	94 (39.0)	26 (10.8)	107 (45.1)	17 (7.2)
Neutrophil count decreased	87 (36.1)	34 (14.1)	79 (33.3)	34 (14.3)
Vomiting	83 (34.4)	7 (2.9)	75 (31.6)	5 (2.1)
Platelet count decreased	82 (34.0)	44 (18.3)	89 (37.6)	38 (16.0)
Aspartate aminotransferase increased	72 (29.9)	0	79 (33.3)	1 (0.4)
Decreased appetite	65 (27.0)	2 (0.8)	59 (24.9)	5 (2.1)
Alanine aminotransferase increased	48 (19.9)	0	47 (19.8)	1 (0.4)
Hypoaesthesia	43 (17.8)	0	37 (15.6)	0
Hypothyroidism	43 (17.8)	0	15 (6.3)	0
Diarrhea	42 (17.4)	3 (1.2)	30 (12.7)	2 (0.8)
Blood bilirubin increased	36 (14.9)	2 (0.8)	33 (13.9)	4 (1.7)
Hepatic function abnormal	29 (12.0)	2 (0.8)	17 (7.2)	1 (0.4)
Weight decreased	27 (11.2)	1 (0.4)	22 (9.3)	1 (0.4)
Gamma-glutamyltransferase increased	25 (10.4)	1 (0.4)	14 (5.9)	3 (1.3)
Asthenia	24 (10.0)	1 (0.4)	17 (7.2)	1 (0.4)
Hypoalbuminemia	24 (10.0)	0	25 (10.5)	1 (0.4)
Hypokalemia	22 (9.1)	3 (1.2)	10 (4.2)	7 (3.0)
Blood alkaline phosphatase increased	22 (9.1)	2 (0.8)	16 (6.8)	1 (0.4)
Thrombocytopenia	19 (7.9)	10 (4.1)	18 (7.6)	8 (3.4)
Lymphocyte count decreased	17 (7.1)	3 (2.1)	15 (6.3)	9 (3.8)
Palmar-plantar erythrodysaesthesia syndrome	17 (7.1)	1 (0.4)	22 (9.3)	3 (1.3)

Abbreviation: CAPOX, capecitabine plus oxaliplatin.

[a] Events are shown in descending order of frequency of grade 1-2 adverse events in the sugemalimab + CAPOX group.

[b] Adverse events were classified according to Medical Dictionary for Regulatory Activities and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.03 (range, 1-5 [1, mild; 2, moderate; 3, severe; 4, life-threatening; and 5, death]). The treatment-related adverse events that led to death in sugemalimab + CAPOX group were pneumonia (n = 1) and hypovolaemic shock (n = 1); the treatment-related adverse events that led to death in placebo + CAPOX group were pulmonary embolism (n = 1) and unknown cause of death (n = 2). For an overview of adverse events, refer to eTable 5 in Supplement 2. For data on potential immune-related adverse events, refer to eTable 6 in Supplement 2.

suggesting that the findings from GEMSTONE-303 may be reasonably extrapolated to broader populations. Second, patients with known *ERBB2* overexpression were excluded. While *ERBB2* testing was not available at all participating sites, patients with unknown *ERBB2* status were included, consistent with the enrollment approach of studies such as CheckMate-649, ORIENT-16, and RATIONALE-305. In GEMSTONE-303, approximately 60% of patients had unknown *ERBB2* status at baseline, evenly distributed between the sugemalimab and placebo groups, minimizing potential bias. Third, microsatellite instability status was not tested, despite evidence that microsatellite instability is associated with improved survival outcomes in patients receiving anti-PD-1 antibodies. Future studies may consider incorporating microsatellite instability testing to refine patient selection and optimize treatment outcomes.

Conclusions

Sugemalimab is the first PD-L1 inhibitor demonstrating superior overall survival and progression-free survival and a manageable safety profile in combination with chemotherapy vs chemotherapy alone in patients with previously untreated locally advanced or metastatic gastric or gastroesophageal junction adenocarcinoma. The results support sugemalimab plus chemotherapy as a new standard first-line treatment option for patients with PD-L1 CPS 5 or greater.

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